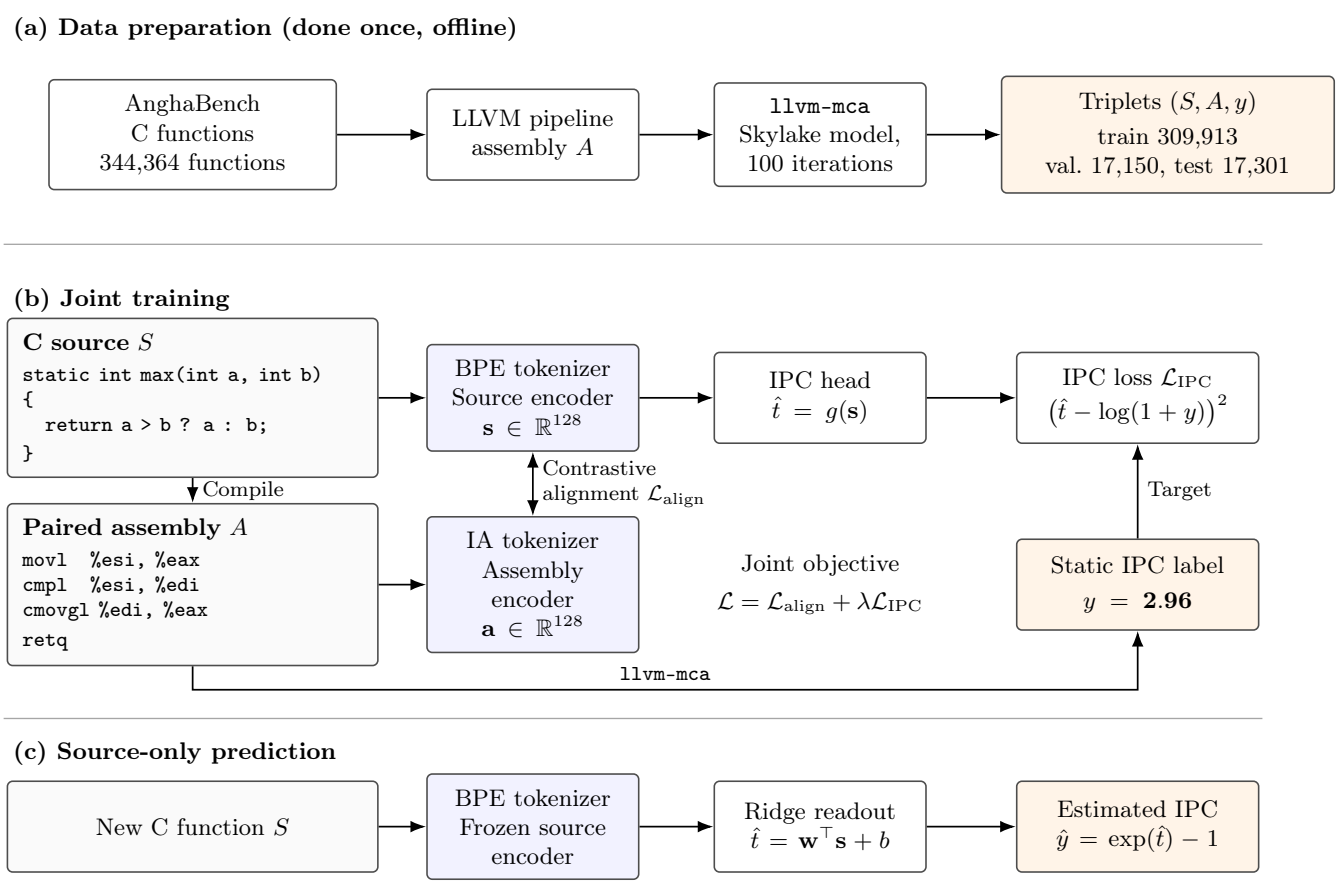

(a) Data preparation (done once, offline)
AnghaBench
C functions
344,364 functions
LLVM pipeline
assembly A
llvm-mca
Skylake model,
100 iterations
Triplets (S, A, y)
train 309,913
val. 17,150, test 17,301
(b) Joint training
C source S
static int max(int a, int b)
{
return a > b ? a : b;
}
Compile
Paired assembly A
movl %esi, %eax
cmpl %esi, %edi
cmovgl %edi, %eax
retq
BPE tokenizer
Source encoder
$\mathbf{s} \in \mathbb{R}^{128}$
Contrastive
alignment $\mathcal{L}_{\text{align}}$
IA tokenizer
Assembly
encoder
$\mathbf{a} \in \mathbb{R}^{128}$
IPC head
$\hat{t} = g(\mathbf{s})$
IPC loss $\mathcal{L}_{\text{IPC}}$
$\left(\hat{t} - \log(1+y)\right)^2$
Target
Static IPC label
$y = \mathbf{2.96}$
Joint objective
$\mathcal{L} = \mathcal{L}_{\text{align}} + \lambda \mathcal{L}_{\text{IPC}}$
llvm-mca
(c) Source-only prediction
New C function S
BPE tokenizer
Frozen source
encoder
Ridge readout
$\hat{t} = \mathbf{w}^\top \mathbf{s} + b$
Estimated IPC
$\hat{y} = \exp(\hat{t}) - 1$